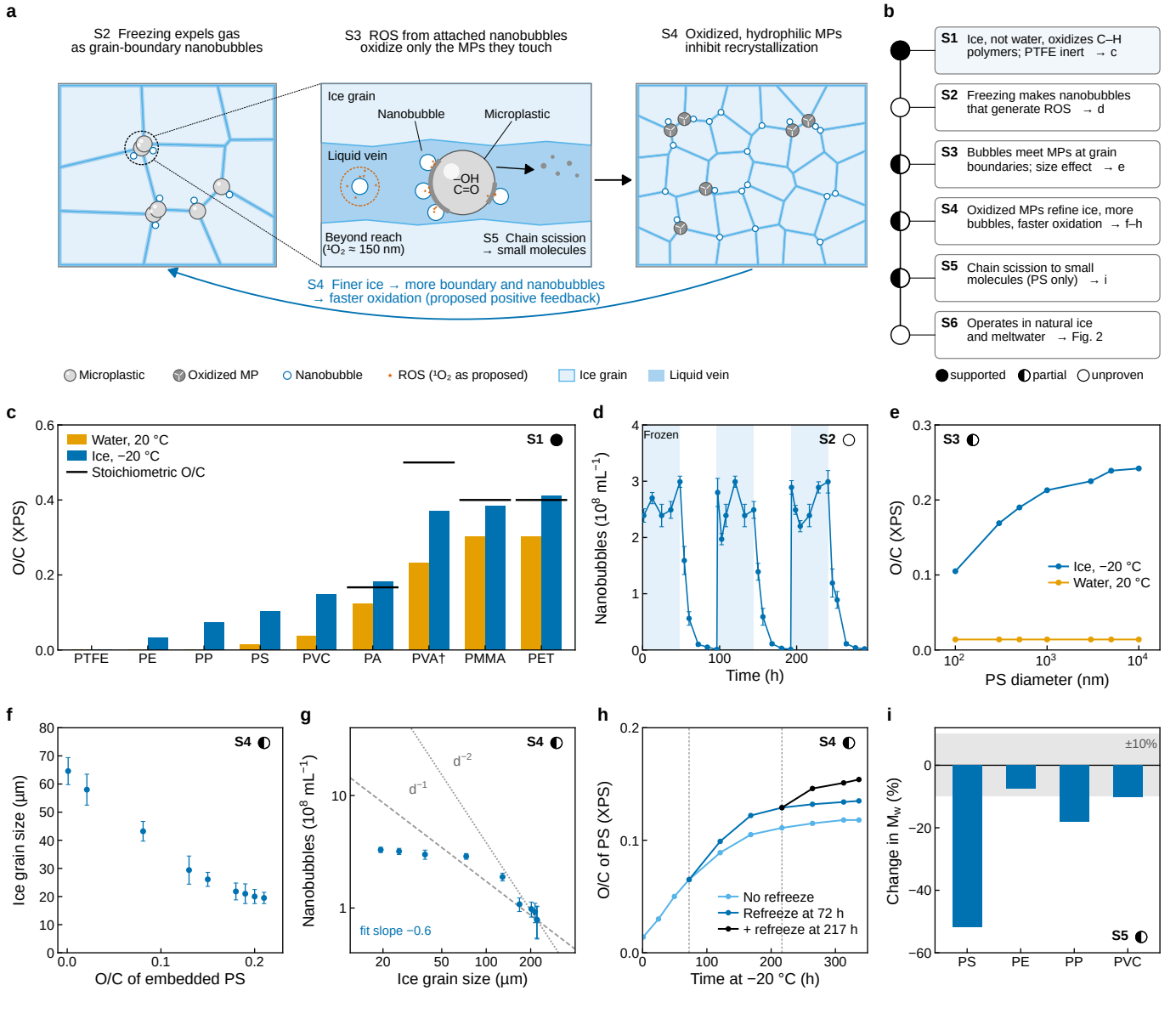

a
S2 Freezing expels gas as grain-boundary nanobubbles
S3 ROS from attached nanobubbles oxidize only the MPs they touch
S4 Oxidized, hydrophilic MPs inhibit recrystallization
Ice grain
Nanobubble
Microplastic
Liquid vein
–OH
C=O
Beyond reach
(¹O₂ ≈ 150 nm)
S5 Chain scission
→ small molecules
S4 Finer ice → more boundary and nanobubbles
→ faster oxidation (proposed positive feedback)
Microplastic
Oxidized MP
Nanobubble
ROS (¹O₂ as proposed)
Ice grain
Liquid vein
b
S1 Ice, not water, oxidizes C–H polymers; PTFE inert → c
S2 Freezing makes nanobubbles that generate ROS → d
S3 Bubbles meet MPs at grain boundaries; size effect → e
S4 Oxidized MPs refine ice, more bubbles, faster oxidation → f–h
S5 Chain scission to small molecules (PS only) → i
S6 Operates in natural ice and meltwater → Fig. 2
supported
partial
unproven
c
Water, 20 °C
Ice, −20 °C
Stoichiometric O/C
S1
O/C (XPS)
PTFE
PE
PP
PS
PVC
PA
PVA†
PMMA
PET
d
Frozen
S2
Nanobubbles (10⁸ mL⁻¹)
Time (h)
e
S3
O/C (XPS)
Ice, −20 °C
Water, 20 °C
PS diameter (nm)
f
S4
Ice grain size (μm)
O/C of embedded PS
g
S4
d⁻²
d⁻¹
fit slope −0.6
Nanobubbles (10⁸ mL⁻¹)
Ice grain size (μm)
h
S4
O/C of PS (XPS)
No refreeze
Refreeze at 72 h
+ refreeze at 217 h
Time at −20 °C (h)
i
±10%
S5
Change in Mw (%)
PS
PE
PP
PVC